

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 1.2 INDICES: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This Final Explanation is your chance to show how everything you learned in Lessons 1 to 8 fits together. Your task is to answer the driving question: How can writing numbers as powers and logarithms help us make sense of, and calculate with, the enormous and tiny quantities that appear in Kenyan farming, finance, engineering, and science? Before you write, look back at your Day-1 model, your notes, your Driving Question Board (DQB) questions, and your work on the Kitale maize doubling puzzle. Answer each section below in complete sentences, in the blank space under the prompt. Use specific evidence from the unit, such as the maize seed sequence 1, 2, 4, 8, ..., 256, the E. coli bacteria, the Richter scale, or your log table work. Do not just state a rule. Explain how and why it works, and show your working for any calculation. Use mathematical notation correctly, for example 2^8 , a^{m+n} and $\log_2 256$.

Section 1: Why write numbers as powers?

A smallholder farmer in Kitale plants 1 maize seed, and the total number of seeds doubles every season. The counts are 1, 2, 4, 8, 16, 32, 64, 128, 256, and by Season 20 the count exceeds 1,000,000. (a) Explain how you can write 256 using index notation, and what the base and the index each tell you. (b) Explain why index notation is more useful than writing out the number or the long string of multiplications when the farmer's numbers become very large. Refer to Season 20 in your answer.

(a) Each season the total is the previous total multiplied by 2, so 256 comes from multiplying 2 by itself eight times: $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^8$. The base, 2, is the number being multiplied repeatedly. The index, 8, tells us how many times the base is used as a factor. (b) By Season 20 the count is $2^{20} = 1,048,576$, which is over one million. Writing that out digit by digit, or as twenty 2s multiplied together, is slow and easy to get wrong, and it would barely fit in the farmer's notebook. Index notation compresses the repeated multiplication into a short expression, 2^{20} , that still shows its structure: what is being multiplied and how many times. This makes it easy to compare seasons (2^8 is clearly smaller than 2^{20}) and to work with the numbers. This is why scientists, engineers and bankers use powers for very large quantities.

Section 2: Why do the laws of indices work?

(a) Using the example $2^3 \times 2^4$, explain in your own words WHY we add the indices when

(a) $2^3 \times 2^4 = (2 \times 2 \times 2) \times (2 \times 2 \times 2 \times 2)$. When we multiply, all the factors of 2 are put together, and we count $3 + 4 = 7$ factors in total, so the answer is 2^7 . We add the indices because we are combining two groups of repeated multiplication into one bigger group. (b) $2^5 \div 2^2 = (2 \times 2 \times 2 \times 2 \times 2) \div (2 \times 2)$. Two factors of 2 cancel from the top and bottom, leaving $2 \times 2 \times 2 = 2^3$. The

multiplying powers of the same base. Expand the powers into repeated factors. (b) Using $2^5 \div 2^2$, explain why we subtract the indices when dividing. (c) Use the Quotient Law to explain why $a^0 = 1$ and why $2^{-3} = 1/8$. (d) Show how the Power Law lets you simplify $(2^3)^2$, and use the laws to work out $2^8 \times 2^4$ as a single power of 2. Explain how these laws connect to the maize doubling puzzle.	number of factors left is $5 - 2 = 3$, so we subtract the indices. (c) Any non-zero number divided by itself is 1. By the Quotient Law, $a^3 \div a^3 = a^{3-3} = a^0$. Both expressions are equal to the same thing, so $a^0 = 1$. This matches the maize table: Season 0 has $2^0 = 1$ seed. To extend further, $2^0 \div 2^3 = 1/8$ by division, and by the Quotient Law it is $2^{0-3} = 2^{-3}$. So $2^{-3} = 1/2^3 = 1/8$, and in general $a^{-n} = 1/a^n$. (d) $(2^3)^2$ means $2^3 \times 2^3 = 2^6$, so the indices multiply: $(a^m)^n = a^{mn}$. Also $2^8 \times 2^4 = 2^{8+4} = 2^{12}$ (which equals $256 \times 16 = 4,096$). These laws turn operations on huge numbers into simple arithmetic on small indices, which is exactly what the maize farmer's notebook needed. They also hold when the index reaches zero or goes negative, because the pattern of multiplying and dividing is unbroken.
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Section 3: From indices to logarithms

(a) In the maize puzzle you know there are 256 seeds. Explain how the question 'how many doublings produced 256 seeds?' leads to a logarithm, and write your answer using log notation. (b) Write $10^3 = 1000$ in logarithm form. (c) Explain why the laws of indices alone could not solve $2^x = 500,000$ when x is not a whole number, and describe what you can say about x using powers of 2 you know. (d) The 2016 Baringo earthquake had a Richter scale reading of 5.9. Explain what this number has to do with indices and why scientists find it useful.	(a) We already know $2^8 = 256$, so the number of doublings is the index 8. The question 'which index of 2 gives 256?' is exactly what a logarithm answers: $\log_2 256 = 8$. A logarithm of N to base a is the index x such that $a^x = N$, so index form and log form are two ways of writing the same relationship. (b) $\log_{10} 1000 = 3$. (c) To solve an index equation with the laws, we write both sides as powers of the same base and then match the indices. 500,000 is not a whole-number power of 2, so we cannot do this. We can only trap x between whole numbers: $2^{18} = 262,144$ is less than 500,000 and $2^{19} = 524,288$ is more, so x is between 18 and 19. Finding x exactly needs a new tool, logarithms. (d) The Richter number 5.9 is an index (a logarithm) attached to a power of 10, so each step up the scale means the shaking is multiplied by a power of 10 rather than just added. Scientists write the small, manageable number 5.9 instead of writing out the huge number 10 to the power 5.9. This is the same idea as the maize puzzle: work with the index, which is a short number, instead of the enormous result.
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Section 4: Calculating with logarithms

A trader needs to find 4680×327 (in KES) without a calculator. (a) Describe the steps you would follow using four-figure common logarithm tables, and name the logarithm	(a) I write each number's logarithm, add the two logarithms (using the law $\log MN = \log M + \log N$), and then find the antilog of the sum to recover the answer. (b) 4680 is between 1000 and 10000, so the characteristic is 3. From the tables, the mantissa is 0.6702, so $\log 4680 = 3.6702$. 327 has characteristic 2 and mantissa 0.5145, so $\log 327 = 2.5145$. Adding: $3.6702 + 2.5145 = 6.1847$. The antilog of the decimal part 0.1847 is about 1.53, and the characteristic 6 shows the order of magnitude, so the answer is $1.53 \times 10^6 = 1,530,000$ (3 s.f.). A calculator check gives 1,530,360, which agrees. (c) Long multiplication of two big numbers involves many partial products where slips are easy. With logarithms, multiplication becomes one addition of two short decimals,
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law you use. (b) Carry out the calculation, showing the characteristic and mantissa of each logarithm, and give your answer to three significant figures. (c) Explain why this method is quicker and less error-prone than long multiplication, and how it links to the Product Law of indices.	plus two table look-ups. The method works because logarithms are indices, and multiplying numbers corresponds to adding indices, just like the Product Law $a^m \times a^n = a^{m+n}$. The characteristic tells us the size (order of magnitude) and the mantissa holds the significant digits.
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Section 5: Answering the driving question	
Now bring your thinking together. Write a connected explanation of the driving question: How can writing numbers as powers and logarithms help us make sense of, and calculate with, the enormous and tiny quantities that appear in Kenyan farming, finance, engineering, and science? Use at least THREE different pieces of evidence from the unit (for example the Kitale maize seeds, E. coli in a Nairobi laboratory, compound interest, SGR or geothermal quantities, or the Richter scale). Explain what indices do, what the laws allow you to do, where the index laws stop being enough, and how logarithms fill that gap. Finish by saying how your thinking has changed since your Day-1 model.	Writing numbers as powers makes enormous and tiny quantities short, readable and comparable. In Kitale, the maize seeds double each season, giving 1, 2, 4, ..., $256 = 2^8$, and by Season 20 the count is $2^{20} = 1,048,576$, over a million. A power keeps the structure: the base shows what is multiplied and the index shows how many times. The laws of indices then let us calculate with these numbers by working with the small indices. When we multiply, indices add ($2^8 \times 2^4 = 2^{12}$); when we divide, they subtract; $(2^3)^2 = 2^6$; $a^0 = 1$; and $a^{-n} = 1/a^n$, which lets us extend the doubling pattern back before Season 0. The same ideas helped with the E. coli bacteria in a Nairobi laboratory, which double every 20 minutes and reach about $2^{30} \approx 10^9$ after 30 doublings, and with engineering and energy quantities written as powers of 10. But the index laws have a boundary: they cannot find x in $2^x = 500,000$, because x is not a whole number (it lies between 18 and 19). Logarithms fill this gap. The logarithm is the index asked backwards: $\log_2 256 = 8$, and $\log_{10} 1000 = 3$. Because logarithms are indices, their laws mirror the index laws: $\log MN = \log M + \log N$, $\log M/N = \log M - \log N$, and $\log M^n = n \log M$. That is why 4680×327 can be done by adding 3.6702 and 2.5145 to get 6.1847 and then taking the antilog to get about 1,530,000. The Richter scale reading of 5.9 for the 2016 Baringo earthquake shows the same compression: scientists quote the short index rather than the huge number. So scientists, engineers and bankers compress huge and tiny quantities into powers and logarithms because this turns slow multiplication into simple addition and keeps numbers manageable. At the start I thought of the maize numbers as just 'big numbers' I could only write out; now I see them as powers I can compare and compute with, and I understand why the Day-1 puzzle is solved.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of index notation and the laws of indices	Explains clearly WHY each law works using expanded factors (e.g. $2^3 \times 2^4 =$	States and applies most laws correctly with some reasoning from repeated	Lists rules with little or no explanation of why they work, or makes errors such as

	2 ⁷), covers the Product, Quotient, Power, Zero and Negative Index laws correctly, and uses notation accurately throughout.	factors; minor notation slips or one law only stated, not explained.	multiplying instead of adding indices.
Understanding of logarithms and their link to indices	Defines a logarithm as the index, converts accurately between index form and log form, and explains why logarithms are needed where the index laws stop (e.g. $2^x = 500,000$), including the Richter scale idea.	Converts between index and log form correctly and gives a basic reason why logarithms are needed; connection to the real-world scale is partial.	Treats logarithms as a separate unrelated topic, or makes conversion errors.
Accuracy of calculation with logarithm tables	Correctly identifies characteristics and mantissas, adds logarithms, finds the antilog, gives an answer close to 1,530,000, and explains the link to the Product Law and why the method is efficient.	Follows the correct method with a small arithmetic or table-reading slip, or gives a weaker explanation of why the method works.	Method is incomplete or incorrect, e.g. multiplies the logarithms instead of adding them, or no working shown.
Use of evidence and connection to the driving question	Uses at least three specific pieces of evidence from the unit (Kitale maize numbers, E. coli, Richter scale, etc.) with correct data, and builds a connected argument that answers the driving question and reflects on growth since the Day-1 model.	Uses two pieces of evidence accurately and addresses the driving question, with limited connection between ideas.	Little or no evidence from the unit, or evidence is inaccurate or not linked to the driving question.

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